

Supplementary Material: A Safety Score Is Not a Certificate

Anonymous submission

Supplementary Evidence Map

The main paper promotes only the shortest evidence path: direct trained-proposer CarCircle action audits, logged episode accumulation, and pre-registered fixed-cap closed-loop audits. This supplement records the completed supporting experiments that do not fit in the seven-page body. All rows use frozen query blocks or pre-registered online audit traces; per-seed CSV files are kept in `analysis/paper_assets/`.

Residual-Horizon Stress

Table 1 changes the logged suffix horizon used to define the residual-cost label on the same CPQ/COptiDICE CarCircle q0.92 protocol. The pattern survives both shorter and longer residual labels: top-selected risk remains nontrivial, and ACCS stays below the 5% target with useful yield.

Logged Episode Proxy and Emission Caps

Table 2 groups fixed action-level query blocks by logged episode. Without an emission cap, small action-level error rates can accumulate into many false issued episodes. Capping the number of certificates per logged episode gives an explicit risk-yield curve.

Closed-Loop Audit Details

Tables 3 and 4 give the audit and test splits behind the main-text closed-loop table. The rule family is fixed before labels are observed: issue only the first m certificates in each online episode for $m \in \{64, 128, 256, 512\}$. The exact one-sided binomial upper bound is reported over issued episodes.

Tune/Audit/Test Closed-Loop Diagnostics

Table 5 records the higher-upside attempt to select a small closed-loop rule family on tune episodes and then freeze the selected rule before audit and test reporting. The broad tuned run reduces CPQ DroneCircle risk relative to the fixed-cap full rule, but does not reach the 5% target. The focused Drone-only follow-ups either issue too few certificates or remain above target. We therefore keep the fixed-cap CarCircle/BallCircle rows as the main positive closed-loop evidence and treat DroneCircle as a hard boundary case.

Broader Environment Action-Level Checks

Table 6 records the one-seed BallCircle/DroneCircle action-level feasibility pass. These rows support portability of the fixed query-bank pipeline but are not stronger headline rows than CarCircle. The follow-up three-seed COptiDICE BallCircle row in Table 7 is more stable, but it is a robustness result rather than a selection-amplification stress row.

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Proposer	H	Cand. F	Top F	Global R/Y	R-bin R/Y	CRC R/Y	ACCS R/Y
COptiDICE	20	6.60	7.35	0.04/13.69	4.01/59.29	2.53/93.83	2.13/93.00
COptiDICE	80	5.63	6.39	5.65/97.94	5.07/95.24	3.83/90.21	3.27/84.93
CPQ	20	5.88	6.24	0.25/8.38	0.89/67.26	2.57/95.26	2.26/94.82
CPQ	80	6.32	7.04	4.11/90.48	1.91/61.03	3.92/86.63	3.61/80.13

Table 1: Residual-horizon stress on official DSRL CarCircle q0.92, averaged over three seeds. Values are percentages. R/Y reports issued false certification / claim yield.

Proposer	Rule	Block R/Y	Episode R/Y	Mean issued blocks
CPQ	ACCS	2.25/80.18	49.08/100.00	33.18
CPQ	ACCS + cap1	0.46/2.42	0.46/100.00	1.00
CPQ	ACCS + cap2	0.57/4.83	1.03/100.00	2.00
CPQ	ACCS + cap4	0.80/9.67	2.53/100.00	4.00
CPQ	ACCS + cap8	1.55/19.33	10.57/100.00	8.00
COptiDICE	ACCS	2.17/78.33	44.37/100.00	32.41
COptiDICE	ACCS + cap1	0.69/2.42	0.69/100.00	1.00
COptiDICE	ACCS + cap2	0.52/4.83	1.03/100.00	2.00
COptiDICE	ACCS + cap4	0.72/9.67	2.64/100.00	4.00
COptiDICE	ACCS + cap8	1.45/19.33	9.54/100.00	8.00

Table 2: Logged episode-block proxy for CPQ/COptiDICE q0.92 on CarCircle. Block R/Y is action-level false issued certification / block yield. Episode R/Y is whether any issued block in a logged episode is false / episode issue rate.

Environment	Proposer	Cap	Audit K/N	Audit U95	Test K/N	Test U95	Issued/ep
CarCircle	CPQ	64	11/600	3.02	12/600	3.22	64.00
CarCircle	CPQ	128	11/600	3.02	12/600	3.22	128.00
CarCircle	CPQ	256	11/600	3.02	12/600	3.22	256.00
CarCircle	CPQ	512	11/600	3.02	12/600	3.22	300.00
CarCircle	COptiDICE	64	370/600	64.97	365/600	64.15	64.00
CarCircle	COptiDICE	128	370/600	64.97	365/600	64.15	128.00
CarCircle	COptiDICE	256	370/600	64.97	365/600	64.15	256.00
CarCircle	COptiDICE	512	370/600	64.97	365/600	64.15	300.00
BallCircle	CPQ	64	0/600	0.50	0/600	0.50	64.00
BallCircle	CPQ	128	0/600	0.50	0/600	0.50	128.00
BallCircle	CPQ	256	0/600	0.50	0/600	0.50	200.00
BallCircle	CPQ	512	0/600	0.50	0/600	0.50	200.00
BallCircle	COptiDICE	64	533/600	90.89	533/600	90.89	64.00
BallCircle	COptiDICE	128	533/600	90.89	533/600	90.89	128.00
BallCircle	COptiDICE	256	533/600	90.89	533/600	90.89	200.00
BallCircle	COptiDICE	512	533/600	90.89	533/600	90.89	200.00

Table 3: Pre-registered fixed-cap saturation audit for the positive CarCircle and BallCircle rows over three 200/200 audit/test seeds per environment-proposer pair. U95 is the exact one-sided 95% binomial upper bound, in percent. Issued/ep is the mean number of issued certificates per online episode; cap512 saturates CarCircle at 300 steps and BallCircle at 200 steps.

Environment	Proposer	Cap	Audit K/N	Audit U95	Test K/N	Test U95	Issued/ep
DroneCircle	CPQ	64	87/600	17.08	90/600	17.61	37.01
DroneCircle	CPQ	128	87/600	17.08	90/600	17.61	43.52
DroneCircle	CPQ	256	87/600	17.08	90/600	17.61	48.12
DroneCircle	CPQ	512	87/600	17.08	90/600	17.61	49.52
DroneCircle	COptiDICE	64	587/600	98.71	576/600	97.23	63.91
DroneCircle	COptiDICE	128	587/600	98.71	576/600	97.23	126.53
DroneCircle	COptiDICE	256	587/600	98.71	576/600	97.23	236.51
DroneCircle	COptiDICE	512	587/600	98.71	576/600	97.23	271.91

Table 4: DroneCircle fixed-cap saturation audit. The shorter issued/ep values for CPQ reflect episodes with far fewer certifiable steps than the nominal cap; the episode-level K/N and exact U95 remain computed over the same three 200/200 audit/test seeds.

Run	Env.	Proposer	Tune K/N	Audit K/N	Audit U95	Test K/N	Test U95
Broad 100/200/200	Car	CPQ	3/300	4/600	1.52	10/600	2.81
Broad 100/200/200	Ball	CPQ	0/300	0/600	0.50	0/600	0.50
Broad 100/200/200	Drone	CPQ	16/197	31/397	10.39	32/395	10.73
Broad 100/200/200	Drone	COpt.	222/230	461/472	98.69	437/455	97.43
Conservative 200/300/300	Drone	CPQ	0/4	4/5	98.98	1/4	75.14
Conservative 200/300/300	Drone	COpt.	119/127	173/186	95.82	153/164	96.19
Medium 200/300/300	Drone	CPQ	22/388	32/581	7.33	54/581	11.52
Medium 200/300/300	Drone	COpt.	242/253	402/420	97.21	366/377	98.36

Table 5: Tune/audit/test selected-rule diagnostics. Tune chooses from a small predeclared rule family; audit/test evaluate the frozen selected rule. U95 is the one-sided 95% binomial upper bound over issued episodes, in percent. These diagnostics are appendix evidence, not a replacement for the fixed-cap main table.

Proposer	Env.	Cand. F	Top F	Global R/Y	R-bin R/Y	CRC R/Y	ACCS R/Y
CPQ	Ball	6.35	5.95	1.72/82.56	1.11/49.51	2.52/90.23	1.88/85.94
CPQ	Drone	2.35	3.04	3.01/99.83	2.82/95.88	2.93/99.97	2.12/88.71
COptiDICE	Ball	6.15	6.53	5.33/96.12	5.27/60.37	3.57/87.64	2.94/80.46
COptiDICE	Drone	2.27	3.19	3.21/100.00	2.98/100.00	3.21/100.00	2.84/98.18

Table 6: Single-seed trained-checkpoint action-level feasibility on official DSRL BallCircle and DroneCircle, q0.92. Values are percentages.

Metric	Mean	Std
Candidate false	5.15	1.42
Top false	4.85	1.59
Global risk	4.03	1.26
Global yield	97.36	1.00
CRC risk	2.82	0.54
CRC yield	91.73	4.87
ACCS risk	2.48	0.33
ACCS yield	88.61	7.19

Table 7: Three-seed COptiDICE BallCircle q0.92 action-level follow-up. Values are percentages.

Off-Policy Query-Shift Diagnostic

Table 8 gives a local diagnostic for the off-policy audit route. The audit law is uniform over a selected-query score, while the deployment law tilts toward high-reward, high-risk queries with density proportional to x^{a-1} . Unweighted auditing estimates the wrong query law; known-ratio weighting and a hard support cap expose the risk-yield and effective-sample-size cost.

Shift	Method	τ	Test R/Y	Audit ESS
4	Unweighted P	0.895	18.65/64.12	17910
4	Weighted Q/P	0.860	4.59/54.67	7426
4	Support cap	0.850	0.00/52.16	17038
4	Oracle Q	0.860	4.59/54.67	109340
8	Unweighted P	0.895	33.90/41.19	17910
8	Weighted Q/P	0.855	4.45/28.49	3902
8	Support cap	0.850	0.00/27.23	17038
8	Oracle Q	0.855	4.45/28.49	56987

Table 8: Off-policy query-shift toy. Test R/Y is deployment false certification / claim yield in percent. The unweighted audit is valid for the audit law P but fails under shifted deployment law Q ; weighting tracks the oracle at the cost of smaller effective sample size, while support capping is conservative.

Diagnostics and Negative Evidence

Diagnostic	Finding	Use in the paper
Exact selection toy	K=16 raises top-selected false certification to 47.91%, while CRC-style and ACCS-v0 repair the toy at about 3.22%.	Phenomenon check, not the main empirical claim.
Policy mismatch	Behavior-continuation residual labels can accept actions that fail under deployment continuation.	Motivates declaring Z_j and separating logged suffix labels from closed-loop labels.
No-overlap diagnostic	Unsupported high-reward regions cannot be certified by distribution-free offline calibration alone.	Motivates support-aware abstention and the no-overlap proposition.
Off-policy shift	Unweighted audit can satisfy the audit-law target while failing under the deployment query law.	Supports the weighted/support-aware extension; not a main DSRL result.
Official HDF5 pilots	Dataset-only and self-generated query pilots verified data paths but produced weak or misleading claim targets.	Kept as reproducibility diagnostics, not promoted.
Weighted BC heads	Multi-head weighted BC ran locally but did not produce a stronger trained-proposer selection gap.	Negative model-search evidence.
SafeFQL comparator	A direct comparison requires runnable checkpoints or code that emits the same fixed query blocks.	Open comparator target, not a completed experimental row.

Table 9: Diagnostic inventory. These runs are useful because they constrain the claim form; they are not all positive headline evidence.